

Liang-Ching Tao

Taiwan | [TODO: email] | [TODO: website] | [TODO: GitHub]

Research Interests

Learning-guided planning; sequential decision-making; reinforcement learning; Monte Carlo Tree Search; neural-guided search; graph-based representations; structured combinatorial optimization.

Education

National Chung Hsing University (NCHU), M.S. in Computer Science and Engineering 2026

Advisor: Prof. Pi-Chung Wang. Graduate GPA: approximately 3.9/4.0 [*needs confirmation: exact GPA*].

Thesis: *A Hierarchical Q-learning and Force-aware Routing Approach for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks*.

National Chengchi University (NCCU), B.S. in Computer Science [TODO: year]

Research Experience

Learning-Guided Planning for Structured Scheduling — Current research project [TODO: dates]

- Develop a system combining a graph-based learned policy prior with MCTS for structured combinatorial scheduling actions.
- Investigate canonical actions, action-space reduction, bounded simulation budgets, and sampled-action search.
- Evaluate learned and handcrafted evaluators through downstream planning utility, reproducible scenarios, and baselines.

Neural-Guided Monte Carlo Tree Search — Research project [TODO: dates]

- Investigated root-level and leaf-level neural guidance and the relation between prediction quality and planning performance.

Hierarchical Reinforcement Learning for Multi-Charger Scheduling — Master's thesis, NCHU [TODO: dates]

- Developed a two-level framework: Q-learning selects regions or scheduling decisions; force-aware routing determines charger movement.
- Modeled coupled charging, routing, and scheduling as a sequential decision problem and built controlled evaluation workflows.

Publications and Manuscripts

- [TODO: exact title]. Third author. WRSN-related work. **Accepted** at [TODO: exact WASN 2026 venue]. [TODO: authors, URL/DOI]
- [TODO: exact title]. First author. MAMM 2026 work. [**Needs confirmation: status**]. [TODO: authors, venue, URL/DOI]
- [TODO: exact title]. First author. MAMM 2026 work. [**Needs confirmation: status**]. [TODO: authors, venue, URL/DOI]

Selected Research Methods

Reinforcement learning and Q-learning; hierarchical decision-making; MCTS; neural search guidance; graph-based policy representations; structured action spaces; scheduling and routing models; baseline and ablation design; reproducible simulation and evaluation.

Research / Teaching Experience

[TODO: RA/TA roles, dates, responsibilities, and institution.]

Honors and Activities

[TODO: awards, scholarships, academic service, workshops, or relevant activities.]